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Geotectonics of the Hanson Lake block, Trans-Hudson orogen, central Canada: a preliminary paleomagnetic report

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Abstract

We present paleomagnetic results from the Sahli granite in the Hanson Lake block of the Trans-Hudson orogen. Evidence shows that the Sahli granite magnetization was acquired during a metamorphic event which probably peaked in the interval 1840–1800 Ma. The reset nature of the Sahli granite data is partly supported by a negative contact test from a cross-cutting mafic dyke. The Sahli granite results and those from the 1844 ± 2 Ma Hanson Lake pluton and Jan Lake granite in the Hanson Lake block indicate high paleolatitudes compared with the low paleolatitudes found for the Wathaman batholith and other rocks of the Slave–Rae–Hearne cratons of similar ages to the north-west of the Trans-Hudson orogen. Estimates of the 'north–south' width of about 4000 km for the Paleoproterozoic Manikewan ocean between the Archean Hearne and Superior cratons using the Hanson Lake pluton and Wathaman batholith poles, for example, are consistent with previous estimates. The angular difference between the Hanson Lake pluton and Sahli granite poles provides an apparent polar wander segment of the geomagnetic field for the Hanson Lake block to compare with the parallel segment given by the poles for the 1851 Ma Reynard Lake pluton and the 1838 Ma Boot Lake–Phantom Lake pluton for the adjacent Flin Flon domain. Both segments indicate a relative north–north-easterly path direction at an apparent polar wander rate within the range 4.3–0.4° per Ma, assuming no large-scale regional tilts of the plutons. The offset between the two apparent polar wander path segments indicates either an 'east–west' component of closure of about 1000 km for the Manikewan ocean or vertical-axis counter-clockwise rotation of units within the Trans-Hudson orogen with respect to the Superior craton plus a possible north–south relative translation between the Hanson Lake block and Flin Flon domain. © 1998 Elsevier Science B.V.

Keywords: Geotectonics; Paleomagnetism; Trans-Hudson orogen

1. Introduction

The Sahli granite is a reworked Archean basement underlying gneiss assemblages in the northern part of the Hanson Lake block of the Trans-Hudson orogen (THO) (Figs. 1 and 2). In an earlier study of the 1844 Ma Hanson Lake pluton to the south of the Sahli granite, it was shown that an ocean basin with a width of about 4000 km existed between the Hanson Lake block and the Archean Slave–Rae–Hearne cratons at about 1844 Ma (Gala et al., 1994). This width is consistent with other paleomagnetic estimates from rock

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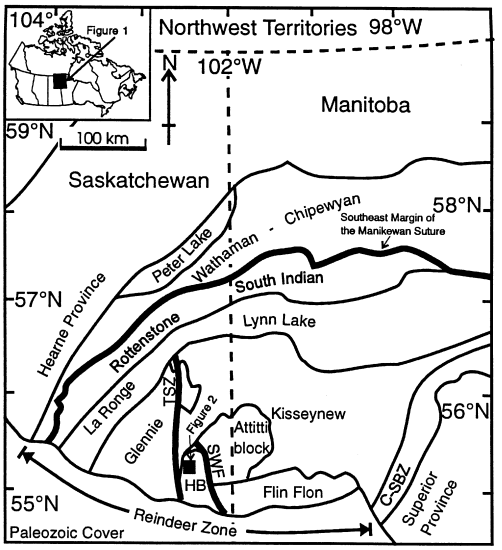


Fig. 1. Location of the study area within the Reindeer zone of the Trans-Hudson orogen. Abbreviations: HB, Hanson Lake block; TSZ, Tabbernor shear zone; SWF, Sturgeon-weir fault; and C-SBC, Churchill–Superior boundary zone.

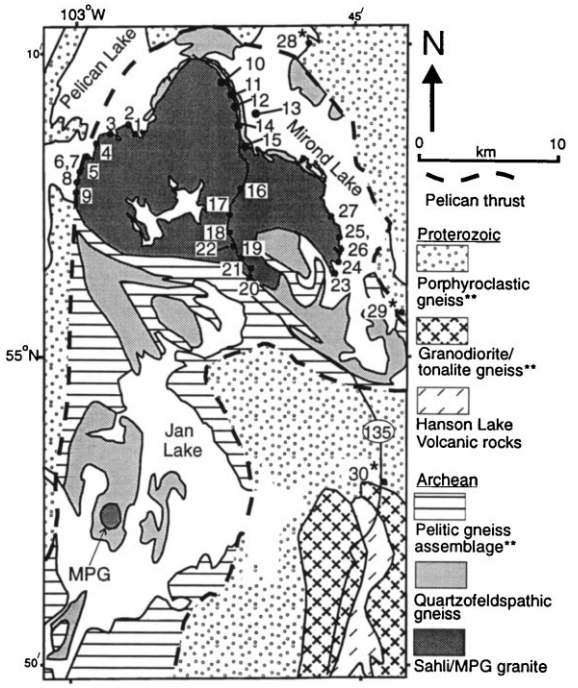


Fig. 2. Simplified geology of part of the Pelican window showing the sampling locations within the Sahli granite. MacMillan Point granite is represented by MPG. The sole zone of the Pelican thrust is represented by thick broken lines. *Gneiss sites. **Jan Lake granite sites are within these map units. Adapted from Lewry et al. (1989) and Lewry (1990).

units within the Reindeer zone (i.e. the internal domains of the THO; Fig. 1; Dunsmore and Symons, 1990; Symons, 1991; Symons et al., 1994. Gala et al. (1994) proposed two models, an 'east–west' translation model and a vertical-axis rotation model, to explain the longitudinal differences between the available 1844 to 1850 Ma poles from within the eastern Reindeer zone. Both models assumed no apparent polar wander (APW) because no APW path segment was available for the Hanson Lake block. The east–west translation model could not account for the pole from the younger 1838 Ma Boot Lake–Phantom Lake pluton (BP) in the adjacent Flin Flon domain (Fig. 1), and the rotation model, in addition to not accounting for the BP pole, also called for an abnormally large angular rotation of the Lynn Lake–La Ronge domain (Fig. 1). Results from the current study are used to improve both models by including some APW and geologic evidence from recent geotectonic models (Ansdell et al., 1995; Hajnal et al., 1996).

The Hanson Lake block has not been previously studied paleomagnetically. Symons (1997) has discussed the rationale for selecting large felsic plutons rather than supracrystal rocks as paleomagnetic targets in the Precambrian orogenic belts such as the THO. In that study, Symons (1997) points out that remagnetization of rocks in an orogen is best avoided by using: (a) felsic rocks because they are much less chemically reactive to hydrothermal solutions than mafic rocks; (b) igneous intrusive rocks because they are less permeable to fluids than jointed porous sedimentary or vesicular volcanic strata; and (c) large batholithic bodies because they act as buttresses around which small plutons and strata are readily deformed. Thus, our studies in the Hanson Lake block have targeted the Hanson Lake pluton, the Jan Lake granite and the charnockitic Sahli granite. However, this rationale does not come without its problems. First, granitic plutonic rocks have weaker magnetization. In this study, this problem was overcome by using a cryogenic magnetometer which is capable of

measuring very weak fields. Second, granitic rocks are generally harder to drill and so sample collection and preparation were difficult. Third, and perhaps most troublesome, the definition of paleohorizontal for plutonic rocks is not without ambiguity (e.g. Symons, 1974; Butler et al., 1989). Some key indicators to expect if a pluton has been tilted would be petrologic differences within the pluton or on opposite sides of its contact aureole. Such differences, if they exist, should be easily detectable when dealing with a large pluton that is many kilometres across because even small tilts should expose large areas of deeper portions of the pluton. No detailed petrologic studies have been performed on the plutonic rocks of the Hanson Lake block, but regional mapping has not revealed any gross petrologic differences which would lead us to suspect tilting of the studied bodies. In fact, it is common for supracrystal rocks around a pluton to be steeply tilted and deformed, but for the pluton to remain subvertical and relatively undeformed. Good examples exist within the western Canadian Cordillera. Thus, in the early assessment of the paleomagnetic history of the THO, in which this study forms a part, the focus has been on large plutonic units with minimal evidence of fabric. In doing so, it should be recognized that the overall pattern of paleomagnetic poles from rock units in the THO is more important than any single pole in interpreting the regional geotectonics because of the possibility of a tilt of the single pole. Further, the plutons selected in the THO, other than the Sahli granite, are amongst the youngest available that show the least evidence of metamorphism and that have been intruded into presumably relatively cool crust. Thus, the ages of magnetizations are likely to be close to the determined U/Pb zircon ages. We, however, wish to emphasize that when dates from other radiometric systems become available a revision may become necessary.

Movement along transcurrent faults may also lead to ambiguities in the interpretation of paleomagnetic results. However, translations of <200 km are usually expected for such faults and this distance falls within paleomagnetic error limitations and so cannot be recovered by paleomagnetic analysis. Thus, most recoverable translations are expected to result from rotations of lithospheric units about distant Euler poles.

2. Geology

The Hanson Lake block lies within the Reindeer zone of the THO (Fig. 1). It is separated from the Glennie domain to the west by the Tabbenor shear zone (TSZ), from the Kisseynew domain to the north-west by the early thrust zone of the Pelican thrust, and from the Flin Flon domain to the south-east and Attitti block of the Kisseynew domain to the north–north-east by the Sturgeon-weir fault (Lewry et al., 1990, 1994; Lucas et al., 1994). The Pelican window is a domal core region in the northern part of the Hanson Lake block. The window exposes Archean Sahli and MacMillan Point granite inliers through the overlying gneisses (Fig. 2; Lewry et al., 1989). Three Archean lithologic assemblages have been identified within the internal suite of the Pelican window (Lewry et al., 1989; Ashton and Shi, 1994). Structurally from lowest to highest, they are the: (1) metamorphic Sahli and MacMillan Point granite inliers; (2) pelitic gneisses; and (3) quartzofeldspathic gneisses. The granites consist of coarse-grained and variably retrogressed granulite-facies gneissic monzocharnockite (Lewry et al., 1989). Ashton and Shi (1994) have reported that peak metamorphic pyroxenes are replaced by aggregates of hornblende, biotite, garnet and magnetite in the Sahli granite. The gneisses of the external suite that envelope the internal suite are synkinematic, highly strained rocks of high metamorphic grade. Pegmatite sheets of the Jan Lake granite complex intrude the external suite to the south of the Sahli granite. Gordon et al. (1990), using U/Pb methods, have determined that peak regional metamorphism in the adjacent Kisseynew domain occurred between 1815 and 1810 Ma. Recent radiometric age dating in the Pelican, Mirond and Jan Lake area (Fig. 2) indicates that the peak of regional metamorphism associated with the overthrusting event occurred in the 1840–1800 Ma time interval (U/Pb zircon, Heaman et al., 1995; L. Heaman, pers. comm., 1996; $^{207}Pb/^{206}Pb$ zircon, Sun et al., 1996). The

Sahli granite and overlying internal suite gneisses have been cut by Archean mafic dykes and sills which have also been metamorphosed. The earliest two phases of deformation in these gneisses, D1 and D2, have been heavily reworked by later deformations, D3 and D4, and so are not easily recognized. The D3 system of northerly trending folds is related with the south-west thrusting over the underlying Archean rocks of the internal suite. The D4 system of north-easterly trending folds, on the other hand, is associated with the initial collision of the Superior craton with the juvenile domains of the THO. The north-easterly trending D5 is the youngest fold system and probably results from protracted collisional effects of the eastern Reindeer zone with the Superior craton.

The collage of lithotectonic domains within the THO were brought together during the closure of the Manikewan ocean (Stauffer, 1984; Fig. 1). The Manikewan suture has been located on the south-east side of the Wathaman–Chipewyan batholith (Symons, 1997; Fig. 1). Here the Archean Slave–Rae–Hearne craton collided with the juvenile terranes of the THO as the Manikewan seafloor subducted beneath the Hearne craton in a north-west sense with terminal collision starting at about 1820 Ma (Ashton and Shi, 1994; Ashton and Lewry, 1994; Lewry, 1994; Symons, 1997). Kinematic indicators in the Hanson Lake block and other parts of the Reindeer zone unequivocally indicate that this major component of the closure was preceded by south-west thrusting of a Paleoproterozoic protocontinent of juvenile arc terranes over the Archean Sask craton, along the Pelican thrust and Sturgeon-weir fault. The Sask craton is exposed today as the Sahli and MacMillan Point granites and the surrounding gneisses of the internal suite in the Pelican window, but is thought to underlie the entire Hanson Lake block and much of the adjacent Glennie domain to the west (Fig. 2; Ashton et al., 1993; Ashton and Shi, 1994; Ashton and Lewry, 1994; Lewry, 1994; Ansdell et al., 1995). The hanging wall of the Pelican thrust includes the Attitti block, the Flin Flon domain and the southern Hanson Lake block. The south-westward thrusting is believed to have lasted from about 1830 to 1805 Ma.

The Sahli granite is exposed over an area of about 100 km^2 (Fig. 2). Bell and Bikerman (1985) obtained Rb/Sr and Nd/Sm ages of 2424 ± 30 and 2420 ± 240 Ma, respectively, that they considered the emplacement age for the Sahli granite. However, Heaman et al. (1995) using U/Pb methods obtained an early metamorphic age of 2450 Ma that they consider to record a metamorphic event at the end of the Archean, suggesting that the earlier ages reflect variable updating and/or mixing. Sun et al. (1996) have obtained a $^{207}Pb/^{206}Pb$ zircon age ~ 2500 Ma, which they also consider to likely represent a metamorphic age. All these ages, however, are at odds with a 1900 Ma age (Bickford et al., 1986) for the granite using the U/Pb zircon method. Paleoproterozoic metamorphism of the Archean inliers which peaked between 1840 and 1800 Ma (Gordon et al., 1990; Heaman et al., 1995; L. Heaman, pers. comm., 1996; Sun et al., 1996) postdate the age of the Sahli granite and overlying metamorphic rocks, and it must predate the unmetamorphosed cross-cutting intrusives like the Jan Lake pegmatite which have been dated to be 1773 ± 9, 1767 ± 1, 1750 ± 20, and in the interval 1784 ± 18–1759 ± 5 Ma (U/Pb zircon, Bickford et al., 1987; U/Pb zircon, Ashton et al., 1992; Rb/Sr whole rock, Bell and Macdonald, 1982; $^{207}Pb/^{206}Pb$ zircon, Sun et al., 1996, respectively).

Metamorphism must also predate deposition of the undeformed and flat-lying Athabasca red sandstones that unconformably overlie the THO over an extensive area to the northwest of the Hanson Lake block. A combination of paleomagnetic and U/Pb ages suggests that these red beds were deposited between 1750 and 1700 Ma (Cumming et al., 1987; Kotzer et al., 1992.

3. Sampling and experimental work

Twenty sampling localities were accessed by boat along the shores of Mirond and Pelican lakes. An additional seven sites were sampled along Highway 135 (Fig. 2). Conventional paleomagnetic methods and tools were used to collect a minimum of five oriented cores of 2.54 cm diameter

at each site. Up to three specimens of 2.2 cm length were cut from each core to give a total of 308 specimens for the collection. Magnetic remanence was measured using a CTF DRM-420 cryogenic magnetometer in a magnetically shielded room at the University of Windsor paleomagnetic laboratory. Thermal demagnetization was performed on one or two specimens from each site in nine steps up to 600°C using a Magnetic Measurements MM1 demagnetizer. A larger number of specimens from a dyke at site 6 and its host rock at site 7 were thermally demagnetized as part of a paleomagnetic contact test. Detailed alternating field (AF) demagnetization was done in a static nontumbling mode in 11 steps up to 110 mT on at least two specimens from each site using a Sapphire Instruments SI-4 demagnetizer. The remaining specimens were AF demagnetized in four steps of 5–20 mT increments up to 60 mT or in five steps of 10–20 mT increments up to 70 mT because the remanence of these specimens tended to decay to the stable end points observed for the specimens that underwent detailed demagnetization. Room-temperature stepwise magnetization of 12 test specimens was carried out up to 900 mT to obtain the saturation isothermal remanent magnetization (SIRM) data. Subsequently, the 12 specimens were AF step demagnetized to obtain the corresponding decay curves.

Characteristic remanent magnetization (ChRM) or end-point directions were obtained using the principal component analysis method of Kirschvink (1980) and, in general, were accepted if the maximum angular deviation was less than 10° over three or more demagnetization steps. An additional eight CHRM directions were obtained for specimens showing incomplete demagnetization using the remagnetization circle method (Bailey and Halls, 1984). Site means were computed using Fisher (1953) statistics.

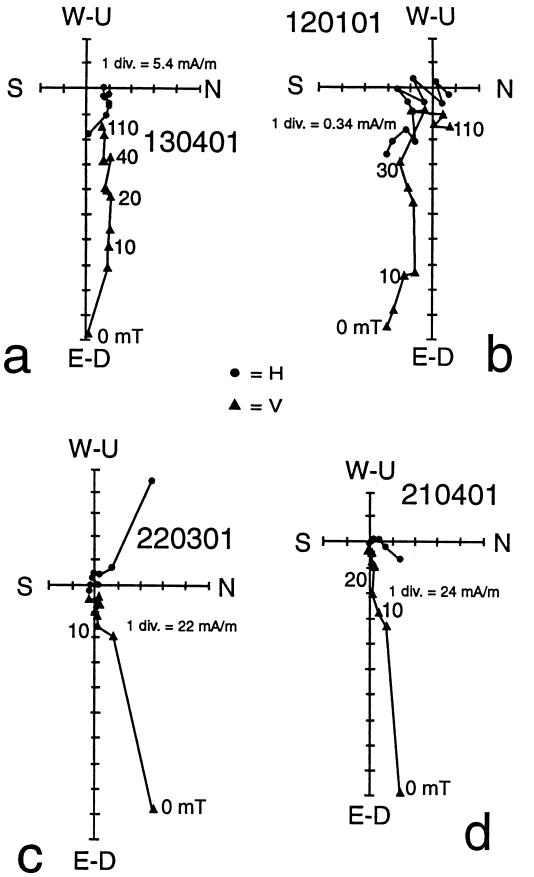


Fig. 3. Typical orthogonal vector diagrams for alternating field (AF) demagnetized specimens from the Sahli granite. The six digit identification numbers give site, core and specimen with two digits assigned to each. Circles and triangles are points on the horizontal (H) and vertical (V) planes, respectively. Abbreviations: N, E, W and S, north, east, west and south; U and D, up and down.

4. Results

The natural remanent magnetization (NRM) intensities range from 10^{-1} to 10^{3} mA/m. Characteristic remanent magnetization was usually isolated between 10 and 70 mT and between 150 and 600°C, and have intensifies that range from 10^{-1} to 10^{2} mA/m. Most ChRM directions recovered after either AF or thermal demagnetization have steeply downward inclinations (Figs. 3 and 4). AF demagnetization shows very rapid decay of the remanence in some specimens [Fig. 3(c) and (d)]. Stepwise thermal demagnetization shows that the ChRM is carried in magnetite [Fig. 4(a) and (b)]. In Fig. 4(c) and (d) are shown two typical examples of thermal demagnetization where titanomagnetite or pure magnetite with distributed grain sizes is the probable carrier of ChRM. In a few cases, pyrrho-

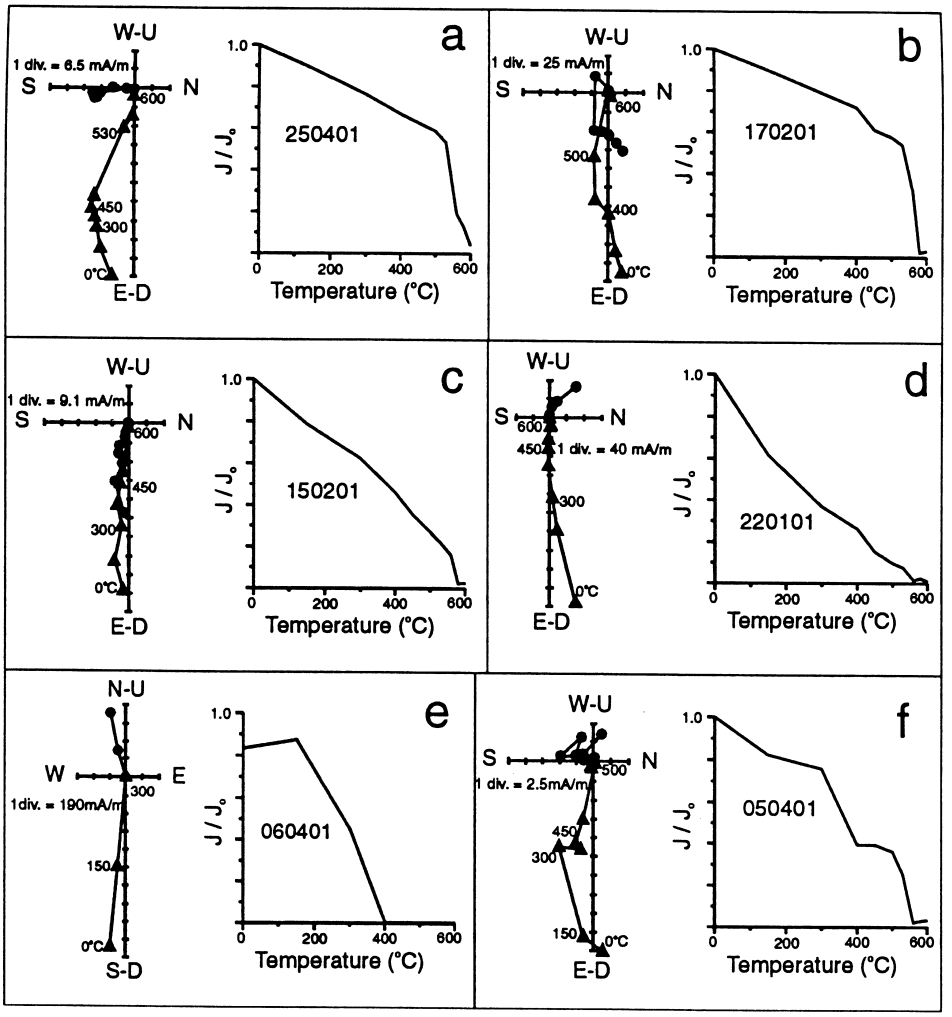


Fig. 4. Selected orthogonal demagnetization and normalized intensity diagrams for thermally demagnetized specimens from the Sahli granite. Conventions are as in Fig. 3.

tite and/or titanomagnetite [Figs. 4(e) and (f)] likely carry the ChRM. The SIRM acquisition curves and the corresponding demagnetization curves for 12 specimens are given in Figs. 5(a) and (b), respectively. This figures, together with Fig. 4(e) and (f), indicate that pseudosingle domain to multidomain magnetite and/or pyrrhotite are the dominant magnetic carriers of the ChRM.

Site 6 is in a mafic dyke site and site 7 is in its adjacent granite host rock. Five dyke specimens give a south-west and moderately steeply upward magnetization [Fig. 6(a)] that is also recovered by thermal and AF demagnetization in six specimens from the dyke's baked contact [Figs. 6(b) and (c)]. Their ChRM directions have been combined to form a 'hybrid' site 6/7. About 10 m from the contact, the magnetization in the six remaining host rock specimens is the steeply inclined and down component [Fig. 6(d)]. They have been retained as site 7 in Table 1. This down direction is also found in five other specimens from the dyke, given as site 6 in Table 1. The implications of these data will be discussed later.

The site mean ChRM directions and the associ-

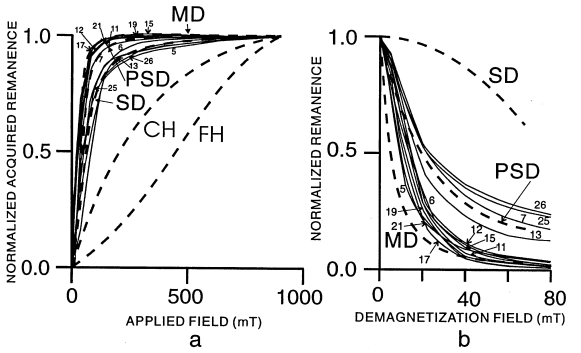


Fig. 5. (a) Saturation isothermal remanent magnetization (SIRM) curves for 12 specimens. Superposed are type curves (thick, broken lines) for single, pseodosingle and multidomain magnetite (SD, PSD, MD) and for coarse and fine-grained haematite (CH, FH) (reconstructed by D.T.A. Symons from Dunlop, 1971, 1972, 1973, 1981); (b) Corresponding SIRM AF demagnetized plots. Small numbers represent site numbers.

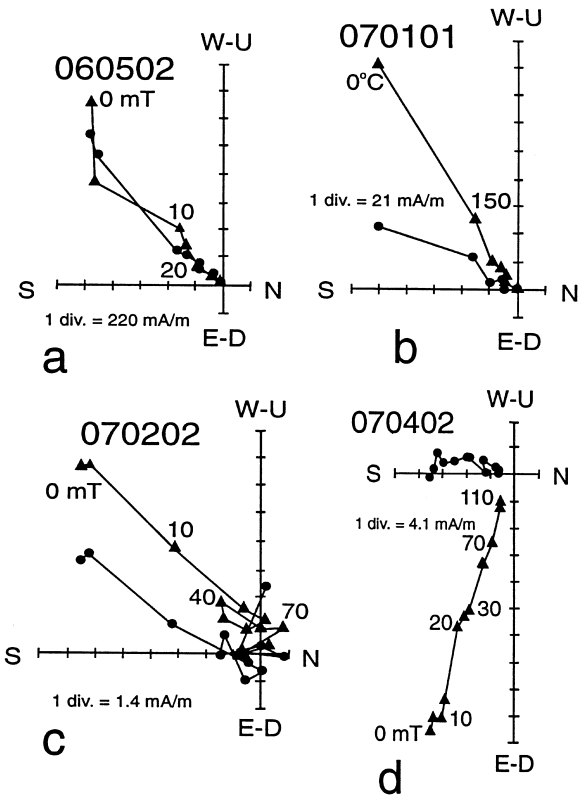


Fig. 6. Orthogonal demagnetization diagrams for: (a) a dyke specimen; (b) and (c) baked contact specimens; and (d) a specimen remote from the intruding dyke. Conventions are as in Fig. 3.

ated statistical indicators are summarized in Table 1. Sites 9, 23 and 27 do not yield homogeneous within site directions, but rather yield two directional modes, at least one of which is deviant and/or poorly defined (Table 1). Also, the mean directions for sites 4, 6, 8, 10 and 14 are poorly defined and the mean directions for sites 6/7 and 18 are deviant. Nineteen of the 23 site means from within the Sahli granite with cones of 95% confidence (α_{95}) of $<15°$ provide a coherent cluster with a unit mean of $D=108.6°$, $I=83.4°$, $\alpha_{95}=4.6$, $k=54.8$, $N=19$ (M, Table 1, Fig. 7). Gneiss site 13 from within the internal suite provides a well-defined mean ChRM direction that is similar to the steep component obtained from the Sahli granite sites (Table 1).

5. ChRM of adjacent gneisses

We have also considered data from three sites in the gneiss sequence of the external suite of the Pelican window from along the east shore of Mirond Lake and along Highway 135 (Fig. 2). They were part of a largely unsuccessful experiment because of remanence instability. In this experiment, samples were collected from 30 sites in amphibolite and granulite gneisses around the window. Detailed AF was carried out on two specimens and thermal demagnetization was carried out on one specimen from each of the three sites. The site mean ChRM directions from the three stably magnetized sites are given in Table 1 as sites 28, 29 and 30 and their overall mean is given as MG. Only these three sites are represented on Fig. 2. Note that gneiss site 13 of the internal suite and the MG gneiss mean are generally similar to the Sahli granite mean (M). This suggests that the Sahli magnetization and all gneiss magnetization were acquired at about the same time and, hence, are likely to be metamorphic remagnetization. It also supports the geologic deduction that the Archean Sahli granite and surrounding gneisses of the internal suite have had the same tectonic history since Archean time (Ashton and Shi, 1994).

Table 1
Site mean ChRM directions

Site	Geology	Number of specimens				Demagnetizing range		Mean remanence vector			
		a	b	c	d	mT	°C	Declination (°)	Inclination (°)	α_{95} (°)	k
1	g	5	10	5	09	10–70	150–530	043.8	64.0	08.2	40
2	g	5	12	4	10	15–60		143.7	80.7	09.2	29
3	g	5	10	3	06	10–50	300–530	114.2	75.9	14.9	21
4[a]	g	5	11	4	07	10–45	400–530	181.6	81.8	15.5	16
5	g	5	10	5	08	10–35	450–530	070.4	82.5	12.3	22
6[a]	md	3	05	3	05	20–50		154.7	76.9	28.6	11
6/7 ~		7	11	7	11	10–110	300–560	208.9	−47.0	11.2	18
7	g	4	09	3	06	10–110	400–600	155.9	73.6	09.6	49
8[a] ~	g	5	12	2	05	15–35	450–600	017.8	48.2	28.4	08
9A[a] ~	g	5	09	3	03	15–30		164.3	59.3	21.2	35
9B[a] ~	g			2	03	15–30	450–600	347.0	31.3	24.4	27
10[a]	g	5	12	2	03	10–30		017.7	77.6	29.1	19
11	g	5	11	5	11	10–50	150–500	166.6	64.4	07.2	41
12	g	5	13	5	13	10–50	450–580	135.1	80.8	05.3	61
13	g	5	13	5	13	05–110	300–600	041.4	71.3	02.8	210
14[a]	g	5	12	4	08	10–30		028.3	85.4	19.5	09
15	g	5	12	4	09	10–40	450–600	169.9	86.8	11.6	21
16	g	5	11	5	08	05–35	450–560	098.5	82.9	08.1	48
17	g	5	12	5	11	10–70	150–530	077.4	80.0	08.6	29
18 ~	g	5	11	4	09	20–110	450–600	353.8	33.5	08.9	34
19	g	5	11	5	11	05–30	150–580	163.3	79.5	05.3	75
20	g	5	11	5	10	15–60	400–580	076.3	83.5	09.2	29
21	g	5	10	5	09	10–50		347.7	82.6	05.8	80
22	g	5	11	4	07	10–50	400–560	229.6	86.4	11.1	30
23A	g	5	11	4	08	20–60	300–580	003.2	84.5	10.5	29
23B ~	g			1	03	25–110		358.6	−24.7	14.2	77
24	g	5	10	5	9	10–50	450–600	083.3	75.8	07.3	50
25	md	5	11	5	11	05–50	150–560	122.7	71.8	09.5	24
26	g	6	13	5	07	10–110	150–530	059.9	85.5	12.2	25
27A	md + g	6	16	4	08	15–45	500–560	005.2	85.7	12.9	20
27B ~	g			2	06	15–45	150–530	301.0	04.6	13.7	25
M					(19)			108.6	83.4	04.6	55
28	gn	3	03	3	03	10–120	000–270	058.0	80.8	16.1	60
29	gn	3	03	3	03	10–120	000–270	136.6	65.0	03.9	1019
30	gn	3	03	3	03	25–120	000–270	160.4	70.3	28.2	20
MG		(9 = 3 + 3 + 3)						134.9	74.9	10.6	25

[a]Site or component means with an α_{95} greater than 15°; these were not used to compute the mean (M). ~ Site or component means that are significantly different from unit direction and so were not used to calculate mean. 'Geol.' is the type of rock sampled (g = granite, gn = gneiss, md = mafic dyke). 'a' and 'b' record the number of cores and specimens measured, and 'c' and 'd' represent the number of cores and specimens used to obtain the site or component mean, respectively. α_{95} and k are cone of 95% confidence limit and precision parameter (Fisher, 1953), respectively. 'M' is the unit mean. Sites 28, 29 and 30 are from a separate collection along the east shore of Mirond Lake (28 and 29) and along HW135 (30). MG is the mean of specimens from sites 28, 29 and 30.

6. Baked contact test

Sites 6 and 7 were collected to provide a paleomagnetic contact test; however, the results are negative. Our thin section observations support the report by Ashton and Shi (1994) that crosscutting dykes and Sahli granite are both metamorphosed and so are very unlikely to provide a positive contact test. Six dyke specimens of site 6 and five baked contact specimens of site 7 give the

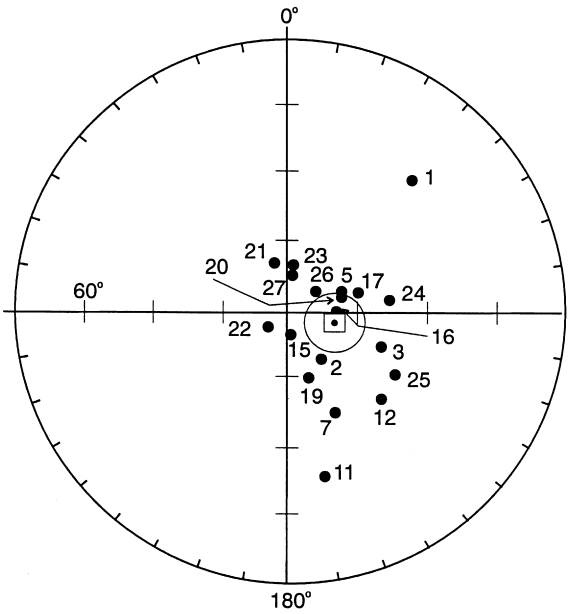


Fig. 7. Equal-area, lower-hemisphere plot of the site mean ChRM directions. The unit mean of the steep and downward component from the 19 acceptable sites is represented by the open square, surrounded by its cone of 95% confidence. Note that the primitive circle represents 50°.

moderately inclined upward ChRM direction. Specimens that are remote from the dyke at site 7 carry the usual host rock ChRM with a steeply inclined downward direction. This, taken on its own, would be a positive contact test that indicates that the dyke magnetization post-dates the Sahli granite magnetization. However, this conclusion is negated by the fact that five dyke specimens give the steeply downward ChRM direction. Closer inspection shows that specimens from hybrid site 6/7 have lower coercivities and higher NRM intensifies than those from sites 6 and 7 with the usual downward ChRM direction (Fig. 6). Therefore, we suggest that the hybrid site 6/7 is recording some post-metamorphic events such as a lightning strikes or contact-controlled fluid flow. The steeply downward ChRM direction of sites 6 and 7 is then the residual metamorphic magnetization. Thus, the Sahli granite and the dyke were emplaced before metamorphism, but both were subsequently remagnetized during a metamorphic event. The question then, is which of the metamorphic episodes (~2420 Ma, and 1840–1800 Ma) was responsible. The answer is quite simple, because both events were high enough to reset magnetization and so the age of the recovered magnetization is likely to fall with the 1840–1800 Ma time interval, during which the second episode peaked.

7. Summary of Hanson Lake pluton and Jan Lake granite results

Recently, Gala et al. (1994) have reported paleomagnetic results from the Hanson Lake pluton. It is an unaltered 1844 ± 2 Ma intrusion to the southwest of the Sahli granite within the Hanson Lake block (Fig. 1). The pluton ranges in composition from pyroxenite to granodiorite and is exposed over an area of about 50 km^2. The paleomagnetic measurements and analyses were performed using the same equipment and methods already described earlier in this paper. AF and thermal step demagnetization provided ChRM directions in 20 of the 23 sites. Of these 20 sites, 16 have good within-site consistency with Fisher (1953) α_{95} values of $<15°$ that give a unit mean ChRM direction of $D=158.6°$, $I=80.0°$, $\alpha_{95}=5.3°$, $k=49$. SIRM acquisition and decay curves indicate that the magnetization is carried by single domain to multidomain magnetite. Recent information suggests that sites 20–23 of the Hanson Lake pluton actually belong to an older body that is marginal to the Hanson Lake pluton. A close examination shows that the site-mean directions from these sites are similar to the directions from the rest of the sites (Gala et al., 1994). Thus, the magnetization in sites 20–23 were likely reset by the intrusion of the Hanson Lake pluton.

The Jan Lake granite refers to a granite–aplite–pegmatite complex that intrudes metamorphosed supracrystal rocks to the south of the Sahli granite (Fig. 2). The granite consists mainly of medium- to coarse-grained pink aplogranite and grey–white, massive to gneissic leucogranite. Pegmatite occurs throughout much of the complex as unfoliated, thin sheets that post-date major ductile deformation. The pegmatite's age is about 1770 Ma (Bickford et al., 1987; Ashton et al., 1992; Sun et al., 1996). Two statistically distinct magnetizations have been recovered for pegmatite

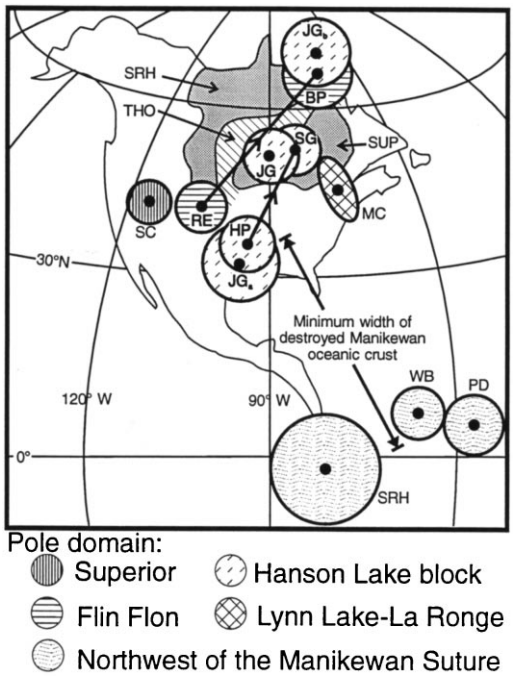


Fig. 8. Pole positions for rocks within the Trans-Hudson orogen and adjacent cratons. Poles from the THO Reindeer zone are Macoun Lake pluton (MC), Hanson Lake pluton (HP), Jan Lake granite (JG), Sahli granite (SG), Reynard Lake pluton (RE), Boot Lake–Phantom Lake pluton (BP), Peter Lake domain (PD) and Wathaman batholith (WB). Poles from the Slave–Rae–Hearne craton (SRH) form a composite pole (SRH) and the reference pole from the Superior craton (SUP) is provided by the Sudbury complex (SC) pole.

site 12 and adjacent gneiss site 13. Magnetization from the remaining sites appear to cluster around these two (Gala, 1996; Gala et al., 1995). It has been inferred that magnetization A (JG_a, Fig. 8) from the gneiss site 13 is probably older than 1844 Ma age of the Hanson Lake pluton, and magnetization B (JG_b, Fig. 8) from the pegmatite site 12 is about 1770 Ma old, the age of the Jan Lake pegmatite. Details of the Jan Lake granite results can be found in Gala (1996) and Gala et al. (1995).

8. Age of the Sahli granite magnetization

The minimum emplacement age of the Sahli granite must be >2420 Ma, the approximate age of early metamorphism. Both the granite and surrounding gneisses are garnetiferous and exhibit high-temperature amphibolite- to granulite-grade metamorphism (Ashton and Shi, 1994) which would destroy its original magnetization. The cross-cutting dyke which we sampled carries a magnetization similar to the Sahli granite magnetization and is itself now an amphibolite dyke. This indicates that both the Sahli granite and the dyke carry metamorphic magnetization. The obvious event that could have caused deep burial of the Sahli granite, and hence resetting of its magnetic signature, is the south-west thrusting of the Paleoproterozoic protocontinent over the Sask craton that hosts the Sahli granite during the 1840–1800 Ma time interval when metamorphism peaked at temperatures that were sufficiently high to reset the U/Pb titanite system (Heaman et al., 1995; L. Heaman, pers. comm., 1996).

Kotzer et al. (1992) have reported several magnetizations that they interpret to record later hydrothermal flushing events in the Athabasca sandstones which unconformably overlie the northern portion of the THO. One is a proposed 900 Ma event, dated by the Rb/Sr method, that gives a magnetization direction that is similar to the Sahli granite direction. This raises the possibility that the Sahli granite magnetization was acquired during this hydrothermal flow event. We note, however, that the Athabasca sandstones are highly porous and, thus, very susceptible to alteration by hydrothermal fluids, whereas the Sahli granite samples are virtually impervious by comparison. Furthermore, there has been no documentation of low-temperature hydrothermal alteration in the Sahli granite. Rather, we see evidence of high temperature recrystallization. Therefore, we reject the possibility of a 900 Ma magnetization and conclude that the best age estimate of the Sahli granite magnetization is within the 1840–1800 Ma interval.

9. Discussion of the Sahli granite pole

9.1. Manikewan ocean ('north–south' component of closure)

The Sahli granite ChRM direction provides a pole (SG) at 49.2°N, 84.0°W, $dp=8.8°$, $dm=9.1°$,

$A_{95}=8.3$ (Fig. 8) and the 1844 Ma Hanson Lake pluton gives a pole (HP) at 36.2°N, 94.5°W, $dp=9.7°$, $dm=10.1°$, $A_{95}=9.5$ (Fig. 8). These indicate that the paleolatitudes of poles for the Hanson Lake block are similar to the paleolatitudes for poles from the 1849 Ma Macoun Lake pluton pole (MC; Symons et al., 1994) in Lynn Lake–La Ronge domain, and the 1851 Ma Reynard Lake and 1838 Ma Boot Lake–Phantom Lake plutons poles (RE, BP; Symons, 1995; MacKay, 1992, respectively) in Flin Flon domain (Figs. 1 and 8). All these domains are to the south-east of the major Manikewan suture between the Rottenstone domain and the Wathaman batholith (Fig. 1). The 1849 Ma Sudbury complex pole (SC; Morris, 1984) from the Superior craton indicates a similar paleolatitude (Fig. 8).

In contrast, poles from the 1855 Ma Wathaman batholith (WB, Fig. 8; Symons, 1991), the ~1855 Ma Peter Lake domain (PD, Fig. 8; Symons, 1994) and the ~1855 Ma composite poles for the Slave–Rae–Hearne cratons (SRH, Fig. 8; see Dunsmore and Symons, 1990 for details) for the north-west of the Manikewan suture have equatorial paleolatitudes. Using the paleolatitudes from the HP and WB poles, for example, we find that terranes to the south-east of the suture were separated from terranes to the north-west by a minimum north–south distance of 4000 ± 1250 km at about 1850 Ma. This provides a minimum estimate of the north–south component of the north-west–south-east closure of the Manikewan ocean.

This paleomagnetic deduction is consistent with the directional sense, but not the magnitudes of transpressional models for the closure of the Paleoproterozoic Manikewan ocean (Gibb et al., 1980; Lewry, 1981; Stauffer, 1984; Hoffman, 1988, 1990). However, our observations do not fit a recent model by Ansdell et al. (1995) which has the Rottenstone domain, Lynn Lake–La Ronge domain and the Glennie-Hanson Lake–Flin Flon 'superdomain' abutting against the Wathaman batholith at 1850 Ma.

9.2. Apparent polar wander trends

The 1844 Ma HP pole is about $17\pm10°$ south-west of the SG pole, giving a north-eastward younging trend for apparent polar wander (APW) in the Hanson Lake block. This APW trend is similar in direction, but smaller in magnitude to that found in the adjacent Flin Flon domain where the poles for the 1851 Ma Reynard Lake pluton (RE; Symons, 1995) and the 1838 Ma Boot Lake–Phantom Lake pluton (BP; MacKay, 1992) have an angular separation of $35\pm15°$ (Fig. 8). The fact that both sets of poles define similar APW trends despite coming from adjacent domains, suggests that they are recording, at least in part, the same APW path. Also, the general south-to-north younging of the APW path between about 1890 and 1849 Ma has been observed both in Lynn Lake–La Ronge domain and the Superior craton (Morris, 1984; Zhai et al., 1994; Symons et al., 1995; Symons, 1997) to the west and east of the Hanson Lake block, respectively. Assuming no relative translation and/or vertical-axis rotation between the Reynard Lake and the Boot Lake–Phantom Lake plutons which are only a few kilometres apart, the 35° angular distance between the RE and BP poles records a $2.7\pm1.1°$ per Ma APW rate. Again, assuming no relative translation and/or vertical-axis rotation and assuming a magnetization age in the interval 1840–1800 Ma for the Sahli granite, the rate of APW for the Hanson Lake block is in the range between $4.3\pm2.5°$ to $0.4\pm0.2°$ per Ma between 1844 and 1840–1800 Ma. The above calculation of APW rate for the Hanson Lake block is probably exaggerated because it does not take into account the fact that prior to the acquisition of the Sahli granite magnetization, the Hanson Lake pluton was part of a bigger unit which moved south-westward with respect to the Archean rocks of the Pelican window (Ashton et al., 1993; Ashton and Shi, 1994; Ashton and Lewry, 1994; Lewry, 1994). It would he possible to correct for this if we had an 1844 Ma old magnetization for the internal suite of the Pelican window. Therefore, at this time, the APW rate for the Hanson Lake block is a first approximation which can be refined if and when sufficient information becomes available. Note, however, that Meert et al. (1993) have documented comparable speeds of more than 16 cm per year for large plates such as Laurentia and Gondwana during the Phanerozoic for time intervals of several tens of

millions of years that will generate rates of APW of $\sim 1.5^{\circ}$ per Ma. Thus, the APW rates we calculate by assuming no relative movements between the rock units in each domain are generally rapid, but certainly tenable.

9.3. Geotectonic models

The two APW path segments obtained for the Hanson Lake block and the Flin Flon domain have an east–west offset (Fig. 8). In addition the 1849 Ma Macoun Lake pluton pole from the Lynn Lake-La Ronge domain (Symons et al., 1994) and the 1849 Ma Sudbury complex pole from the Superior craton (Morris, 1984) are also offset in an east-west sense from the nearly coeval HP and RE poles from the Hanson Lake block and Flin Flon domain (Fig. 8). In order to resolve these discordances, we consider two fundamental types of tectonic motion, namely vertical-axis rotation and translation which is a rotation about a distant Euler pole. However, before doing that, we address two issues.

First, the average of the poles from the 1850 ± 3 Ma norite and 1849 ± 3 Ma granophyre of the Sudbury complex (SC, Morris, 1984) is plotted on Fig. 8. Morris (1984) concluded that the magnetization recovered from the norite is a metamorphic overprint related to the granophyre intrusion and so the best estimate of the age of the nolite magnetization is 1849 ± 3 Ma. To calculate the pole, Morris (1984) applied differential tilt-corrections for the north, east, south and west ranges of the complex in order to account for the slumping of the Sudbury basin with the north range directions receiving a much smaller amount of tilt-correction than the south range. Seismic reflection imaging (Milkereit et al., 1992) defines gentle dips for the north and much steeper dips for the south range in accord with the correction used by Morris (1984). Thus, we believe that the Sudbury complex pole reported by him is a true record of the ambient field for the Superior craton at about 1849 Ma and that it can be used in calculating domain offsets from paleopoles in the THO south-east of the Manikewan suture.

Second, we shall also assume that tilt-corrections for all the three units from the Hanson Lake block are not necessary. The definition of paleohorizontal for plutonic rocks is not without ambiguity. However, we do know that the Pelican window is a gentle and subcircular domain. In the Hanson Lake pluton, Jan Lake granite and Sahli granite, we also find geographic consistency of the ChRM directions within large sampling areas. Perhaps even more compelling is the fact that continuous low-angle thrust and sole faults are imaged under the Hanson Lake block and Flin Flon domain by 'LITHOPROBE' seismic profiles (Lucas et al., 1993; Hajnal et al., 1996). These are accepted as evidence that the Sahli granite, Jan Lake granite and Hanson Lake pluton have not been significantly tilted about subhorizontal axes since acquiring their ChRM. This assumption of no tilt-correction will have to be reassessed when direct evidence of tilt such as differential degrees of deformation around the boundaries of the plutons or variations in petrologic phases with geobarometric import within the plutons become available.

9.3.1. Nonconcurrence of vertical-axis rotation models

Vertical-axis rotation needs to be considered because, although the pole positions may be statistically distinct, the paleolatitudes of the poles that they record may not be different (that is they differ, principally in paleolongitude). This is true for the domains on the south-eastern side of the Manikewan suture.

Two kinds of vertical-axis rotation models are possible: (1) local rotation of units within their tectonic blocks; and (2) regional rotation of terranes like the Flin Flon domain or the Hanson Lake block with respect to one another. An intrinsic assumption of vertical-axis rotation models is that the tectonic units or terranes have not been displaced relative to one another. Under model 1, we find that a 27° clockwise rotation of the Reynard Lake pluton pole brings it into agreement with a fixed Sudbury Complex pole. This calls for 27° counter-clockwise rotation of the Reynard Lake pluton relative to the Superior craton subsequent to 1851 Ma. Similarly, the discordance between the Hanson Lake pluton, Jan Lake granite A poles, and Sudbury Complex pole call for 58° and 48° counter-clockwise rotation of the Hanson

Lake pluton and Jan Lake granite with respect to the Superior craton, respectively. Due to lack of a pole in the 1840–1800 Ma interval from the Superior craton, we cannot propose similar corrections for the units carrying younger magnetization from the Hanson Lake block and Flin Flon domain. Under model 2 and using the Reynard Lake pluton and Hanson Lake pluton poles referenced to the Sudbury complex pole, counter-clockwise rotations of the Hanson Lake block and Flin Flon domain by 58° and 27°, respectively, are required with respect to the Superior craton to account for the discordance between the three poles.

Differential vertical-axes rotations of units within orogenic belts are a common feature. Irving and Wynne (1990, 1991), for example, have provided summaries of similarly inferred terranic rotations in the Canadian Cordillera. Nevertheless, model 1 is not supported by geologic information. In the Flin Flon domain, the Reynard Lake and Boot Lake–Phantom Lake plutons lie side by side and have similar regional trends at the sampled areas. It is, therefore, unlikely that these units have been rotated about vertical axes relative to one another. In the Hanson Lake block, the Hanson Lake pluton and the Jan Lake granite both intrude rocks of the Amisk group (or equivalents) with similar north to south regional trends. These trends change very slightly to north–north-east to south–south-east as we enter into the Sahli granite. Thus, geologic evidence suggests no rotation possible between the Hanson Lake pluton and the Jan Lake granite and that together they have undergone little or no rotation relative to the Sahli granite. Also, if model 1 were true then the paleolatitudes of the sampling sites, which today are all within a radius of less than 1° on the Earth's surface, should be the same. Because of large errors introduced due to variability in the declinations of our steep directions, it is hard to decide whether the sampling site paleolatitudes differ or not. We, therefore, restrict ourselves to the geologic evidence and, hence, discount model 1.

Following model 2, rotation of the Hanson Lake block and Flin Flon domain to Hanson Lake pluton and Reynard Lake pluton poles into agreement with the Sudbury complex pole has two consequences which are hard to justify. First, it results in the APW path segments becoming more divergent, suggesting that the magnitudes of the vertical-axes rotations of the terranes are too high. This conflicts with regional geologic trends in the sampled localities of the Hanson Lake block and the Flin Flon domain which differ slightly by less than 10°. Second, it requires that the APW trend for the THO swings fortuitously from north–north-east to east–north-east at about 1850 Ma, thereby skewing the general north–south trend (Symons et al., 1995). Although possible, there is no evidence for or against such a convenient coincidence at this time. Thus, it is concluded that model 2 rotations are possible, but unlikely.

9.3.2. Translational model

In this model, we appeal to rotations about distant Euler poles. Unlike the local vertical-axis rotation model, we assume in this case that the tectonic terranes have been displaced significantly (that is >500 km) relative to one another since the magnetizations were acquired. Thus, minor displacements along local faults are unlikely to be recovered. In addition, whatever displacement we apply to the Flin Flon domain relative to the Superior craton in order to account for the 1851 Ma Reynard Lake pluton pole must also be applied to account for the Boot Lake–Phantom Lake pluton pole. Also, similar displacements will have to be given to the 1844 Ma Hanson Lake pluton, Sahli granite and Jan Lake granite A poles when the Hanson Lake block is translated away from the Flin Flon domain.

First, we consider the longitudinal differences. Keeping the Sudbury complex pole fixed and using its longitude for reference, we find that bringing the Macoun Lake pluton, Hanson Lake pluton and Reynard Lake pluton poles into coincidence requires the westward translation of the Lynn Lake–La Ronge domain, the Hanson Lake block and the Flin Flon domain away from the Superior craton by 3800 ± 1040, 1800 ± 1250 and 730 ± 1220 km, respectively [Figs. 9(a) and (b)]. A very important observation about these distances is that the farther west a terrane is today from the Superior craton, the larger the required westward translation of the terrane. Thus, the east–west

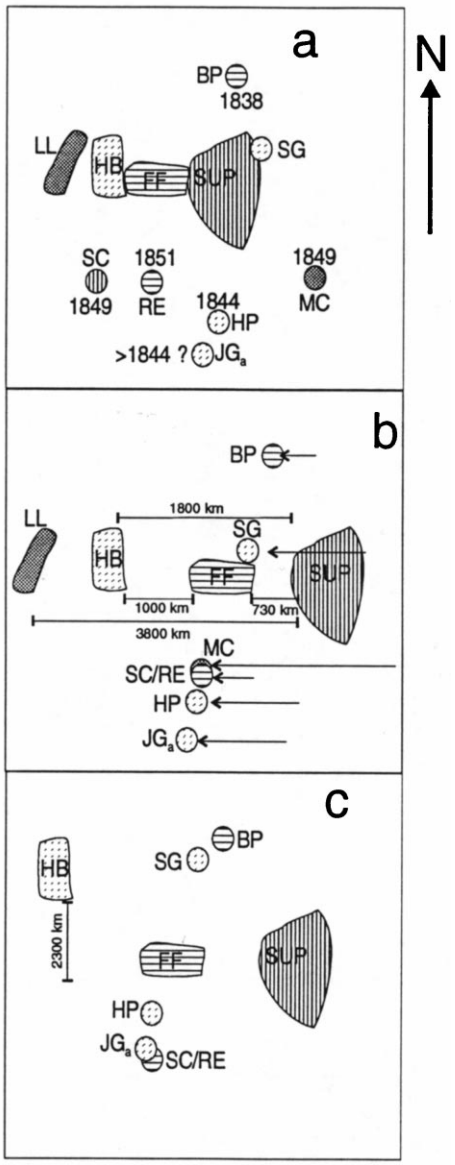


Fig. 9. Schematic representation of: (a) the MC, HP, JG_a, SG, RE, BP and SC poles, the Lynn Lake–La Ronge domain (LL), Hanson Lake block (HB), the Fhn Flon domain (FF) and the Superior craton (SUP) in today's coordinates; (b) the MC, HP, JG_a, SG, RE, BP poles and LL, HB, FF after correction for east–west translation so that RE, HP, and MC now coincide with the longitude of the SC pole. Lengths of arrows are proportional to the relative translation; and (c) the HP, JG_a and SG poles and HB after correction for north–south translation.

component, together with the north–south component noted earlier in this paper, is clearly consistent with the north-west–south-east closure of the Manikewan ocean. Evidence abounds in the THO for a terminal collision of this sense that closed these intervening ocean 'basins'. For example, the pattern of major shear zones suggests indentation by the Superior craton with the Kisseynew domain, Hanson Lake block and Flin Flon domain collectively being the squeezed intervening wedge. Also, the Tabbernor fault is a major dextral shear zone along which material at the apex of the wedge was squeezed to a greater extent (Gibb, 1983). A recent model by Ansdell et al. (1995) calls for continued compression to as late as 1810 Ma. If we now assume that the east–west separation between the Lynn Lake–La Ronge domain, the westernmost domain from the Superior craton that has been considered, and the Superior craton represents the 3800 ± 1040 km east–west width for the Manikewan ocean, then it is possible to now combine this information with the previously described 4000 ± 1250 km north–south width to compute a minimum 'north-west–south-east' width for the Manikewan ocean. This is found to be about 5500 km at a bearing of 317° in today's geographic reference frame.

More specifically, for example, these data indicate that an ocean basin with an east–west width of about 1000 km existed between the Hanson Lake block and Flin Flon domain within the time interval from $\sim$1850 to $\sim$1800 Ma. Note, however, that this does not contradict the fact that prior to 1830 Ma, most of the Hanson Lake block and the Flin Flon domain were part of the Paleoproterozoic protocontinent on the Manikewan seafloor which was moving south-westerly relative to the Sask craton and was thrust over the craton, as seen in the Pelican window. Rather, the 1000 km wide basin was part of the protocontinent transported south-westward and then terminally closed after 1830 Ma.

Next, we consider the north–south ordering of latitudes of poles for the Hanson Lake block and Flin Flon domain. After correction for the east–west component of closure, the 1844 Ma Hanson Lake pluton pole plots to the south of the 1851 Ma Reynard Lake pluton pole [Fig. 9(b)], thereby

violating the south-to-north age progression seen overall in the THO and adjacent Superior craton. Making the assumption that cooling rates subsequent to isotopic blocking were the same in all units, a case can be made for an additional relative north–south translation between the Hanson Lake block and Flin Flon domain. We have already calculated an APW rate for the Flin Flon domain to be 2.7° per Ma between 1851 and 1838 Ma. Also, we obtain an APW rate of between 4.3° and 0.4° per Ma between 1844 and 1840–1800 Ma for the Hanson Lake block. If we now take the arithmetic average rate of ~2.4° per Ma for the Hanson Lake block and assume that the same seafloor spreading velocity applies to both the Hanson Lake block and the Flin Flon domain, then the arithmetic average APW rate for the two domains is ~2.5° per Ma. Applying this average APW rate to the 1844 Ma (94.5°W, 36.2°N, $dp=9.7°$, $dm=10.1°$) Hanson Lake pluton and the 1851 Ma (106.8°W, 39.7°N, $dp=9.2°$, $dm=9.4°$) Reynard Lake pluton poles suggests that the Hanson Lake block was located north of the Flin Flon domain by $21\pm11°$ (~2300 ± 1200 km) [Fig. 9(c)]. This, combined with the 1000 km east–west component between the two domains, defines a north-west–south-east separation of ~2500 km between Flin Flon domain and Hanson Lake block at ~1844 Ma with subsequent compressive and/or transpressive closure.

10. Conclusion

The conclusions of this study have varying degrees of certainty.

(1) The north–south width of ~4000 km for the Manikewan ocean is supported by a large body of paleomagnetic data and it fits conceptually with the generally accepted north-west–south-east closure model for the ocean. We believe that this estimate is now well established.

(2) The magnitude of the east–west component of closure is calculated to be in the order of 3800 ± 1040 km with potentially large errors. Obviously, however, the north-west–south-east closure model for the Manikewan ocean must have an east–west component and, thus, we believe that our paleomagnetic results provide a reasonable initial estimate for it. This conclusion, although consistent with the closure model for the Manikewan ocean, is subject to review if and when evidence of tilt becomes available.

(3) The possibility of a relative north–south or sinistral displacement of 2300 ± 1200 km between the Hanson Lake block and Flin Flon domain is based on small differences in ages between the Reynard Lake, Boot Lake–Phantom Lake, and Hanson Lake plutons, and the Sahli granite magnetization, coupled with the assumption that the magnetization correlate exactly with the mostly U/Pb zircon ages. Thus, this estimate is correct in sense, but somewhat speculative in magnitude at this time, since the U/Pb system closes at temperatures higher than the blocking temperatures of thermoremanent magnetization (TRM). When cooling rates and/or ages from radiometric methods that are more representative of TRM blocking temperatures become available, revision may be done.

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